Q5 – List Comprehension: Cleaning Sensor Readings

Cleaning

Retain readings in the inclusive sensor range –50°C to 50°C. Convert valid Celsius values to Fahrenheit with F=C×9/5+32. enumerate supplies each minute index; alarm minutes are readings outside the safe 2–8°C range.

Comprehensions

A list comprehension has the form [expression for item in iterable if condition]. Use a loop when the logic has multiple branches or side effects. A dictionary comprehension maps minute to temperature; a set comprehension extracts distinct alarm temperatures. A generator expression produces values lazily and generally uses less memory than materialising a list.

Complexity

Cleaning and conversion are O(n) time. The output lists require O(n) space; a generator can reduce intermediate storage if results are consumed once.

Python Program

```
readings = [3.2, 7.1, 51.0, -55.0, 4.5, 1.2, 8.4]
valid = [x for x in readings if -50 <= x <= 50]
fahrenheit = [round(x * 9/5 + 32, 2) for x in valid]
alarms = [(minute, temp) for minute, temp in enumerate(readings)
          if not 2 <= temp <= 8]
print("valid C:", valid)
print("valid F:", fahrenheit)
print("alarm minutes:", alarms)
```